

Enabling grid services by water distribution systems: Measurement-based modeling and optimized scheduling

Stelios Dimoulas^{*} , Georgios Kryonidis , Kyriaki-Nefeli Malamaki , Pandelis Biskas

School of Electrical and Computer Engineering, Aristotle University of Thessaloniki, Thessaloniki, Greece

HIGHLIGHTS

- Simplified, measurement-based modeling approach for water distribution systems (WDSs).
- WDS optimization for interruptible/curtailable load and balancing service provision.
- Two-stage (day-ahead and intra-day) convex optimization, including service activation.
- Application to a real-world WDS in Greece.
- WDSs are suitable for grid service provision, offering significant economic benefits.

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ABSTRACT

Water distribution systems (WDSs) can be optimally controlled and scheduled to provide grid services, such as interruptible/curtailable load and balancing services, to the electric power system, by exploiting the flexibility of water pumps and the storage capacity of water tanks. Nonetheless, existing frameworks for modeling and optimally scheduling WDSs rely on detailed parameter knowledge, involve complex mathematical formulations, and do not account for all time horizons associated with the provision of grid services. To address these issues, this paper presents a practical two-fold methodology for enabling WDSs to provide multiple grid services. First, a new measurement-based modeling approach for WDSs is proposed to eliminate the need for parameter knowledge. The approach correlates the electric power consumption and flow rate of pumps via a convex quadratic function, enabling simpler yet more accurate modeling. Second, by leveraging this modeling approach, a comprehensive two-stage optimization framework is developed to derive the WDS operating schedule, accommodating multiple grid services across both the day-ahead and the intra-day time horizons. The proposed methodology is applied and validated on a real-world WDS in Greece. The results demonstrate the accuracy of the proposed modeling approach, the effectiveness of the proposed optimization framework, and the suitability of WDSs as flexible loads for grid services.

1. Introduction

1.1. Motivation

The controllability of water pumps and the storage capacity of water tanks render municipal water distribution systems (WDSs) highly flexible loads [1,2] within the electric power system (EPS). In the context of the water-energy nexus and integrated energy systems, this flexibility enables WDSs to participate in both implicit and explicit demand-side flexibility schemes [3], allowing them to provide operational

benefits to the EPS while receiving economic benefits in return. Implicit demand-side flexibility schemes include dynamic electricity pricing programs, whereas explicit demand-side flexibility schemes involve the provision of grid services to the EPS [1], such as interruptible/curtailable load (I/C-L) [4,5] and balancing services [6].

In this context, the participation of WDSs in demand-side flexibility schemes constitutes an optimal scheduling problem. Specifically, the WDS operating schedule is optimized to minimize the net operating cost, based on the structure of the flexibility schemes involved. When providing grid services, the associated remuneration is also considered.

^{*} Corresponding author.

Email addresses: steliosd@ece.auth.gr (S. Dimoulas), kryonidi@ece.auth.gr (G. Kryonidis), kyriaki_nefeli@hotmail.com (K.-N. Malamaki), pbiskas@auth.gr (P. Biskas).

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Nomenclature of the Optimization Framework

Sets

$\mathcal{N}_p$	Set of pumps.
$\mathcal{N}_\tau$	Set of tanks.
$\mathcal{N}_{in,\tau}$	Set of pumps upstream of tank τ .
$\mathcal{N}_{out,\tau}$	Set of pumps downstream of tank τ .
$\mathcal{T}$	Daily (day-ahead) time horizon.
$\mathcal{T}_{I/C-L}$	Time period for the provision of the interruptible/curtailable load (I/C-L) service.
$\mathcal{T}_R$	Time period for the provision of the balancing service.
$\mathcal{T}_{id}$	Time horizon of the intra-day problem.

Parameters

Δt	Analysis time step.
t_0, t_f	Initial and final time intervals of $\mathcal{T}$, respectively.
$C[t]$	Electricity price.
$C_{I/C-L}[t]$	Compensation rate for I/C-L service.
$C_u[t]$	Compensation rate for upward reserve capacity.
$C_d[t]$	Compensation rate for downward reserve capacity.
$P_b[t]$	Baseline power consumption of the WDS.
$Q_\tau^c[t]$	Forecast aggregate water demand to be supplied by tank τ .
$P_p^{\max}, P_p^{\min}$	Operating maximum and minimum electric power consumption of pump p .
$Q_p^{\max}, Q_p^{\min}$	Operating maximum and minimum flow rate of pump p .
$SoC_\tau^{\max}, SoC_\tau^{\min}$	Operating maximum and minimum state-of-charge (SoC) limits of tank τ .
V_τ	Volume of tank τ .
SoC_τ^0	Initial SoC value of tank τ at t_0 .

t_a	Time interval for activation of the balancing service in the intra-day stage.
R^a	Balancing power requested by the electric power system operator for the balancing service activation at t_a .
$P_{s,a}^{DA}$	Total WDS power consumption at t_a according to the day-ahead schedule.
SoC_τ^a	SoC value of tank τ at t_a .
c_τ	Numerical weight to ensure SoC recovery for tank τ at the intra-day stage.

Decision variables

$P_p[t]$	Continuous variable modeling the electric power consumption of pump p .
$Q_p[t]$	Continuous variable modeling the flow rate of pump p .
$k_p[t]$	Binary variable modeling the shutdown of pump p (on/off status).
$P_p^k[t]$	Mixed discrete-continuous variable modeling the electric power consumption of pump p including shutdown.
$Q_p^k[t]$	Mixed discrete-continuous variable modeling the flow rate of pump p including shutdown.
$SoC_\tau[t]$	SoC of tank τ .
$SoC_\tau^u[t]$	Auxiliary variable for the SoC of tank τ in case of upward reserve activation.
$SoC_\tau^d[t]$	Auxiliary variable for the SoC of tank τ in case of downward reserve activation.
$P_{I/C-L}[t]$	Power reduction capacity of the WDS for the I/C-L service.
$R_u[t], R_d[t]$	Upward and downward reserve capacities of the WDS for the balancing service.

Furthermore, the optimization problem employs a WDS hydraulic model to ensure that technical requirements related to WDS operation are satisfied.

In this regard, the relevant literature exhibits gaps and shortcomings related to both the accommodation of demand-side flexibility schemes and the hydraulic modeling of WDSs within the optimization problem.

1.2. Literature review: demand-side flexibility schemes

The participation of WDSs in dynamic electricity pricing programs has been thoroughly addressed in the scientific literature [7], with studies focusing on both fixed-speed pumps (FSPs) [8–10] and variable-speed pumps (VSPs) [11–14], [15]. Nevertheless, the challenge of providing multiple grid services to the EPS has not been adequately addressed, as most studies focus exclusively on either I/C-L [16–19] or balancing services [20,21].

Specifically, initial studies on the I/C-L capability of WDSs focused solely on FSPs [16–18], thereby substantially limiting controllability and flexibility. In [19], the participation of WDSs in I/C-L schemes of the French market is examined. Although the market structure is thoroughly modeled, the physical modeling of the WDS, and especially of the pumps, is not detailed.

Regarding balancing services, they involve two stages: the availability stage and the activation stage. The availability stage pertains to the reservation of upward and downward balancing capacities through day-ahead scheduling, while the activation stage concerns the actual delivery of these capacities in the intra-day horizon upon request by the EPS operator. Nevertheless, the existing studies do not comprehensively cover both stages. To elaborate, in [20], the authors explore the ability of WDSs to provide downward-only balancing services by flexibly increasing their power consumption to absorb surplus renewable energy. However, this study focuses solely on the activation stage of the service, as the capacity to increase the WDS power consumption is assumed to be known in advance. In [21], the provision of short-term

balancing services is examined. Nonetheless, by considering a single, symmetric balancing capacity for both upward and downward service provision, and quantifying it on a per-pump basis rather than system-wide, the study underestimates the capability of the WDS to provide balancing services. In addition, [21] does not optimize the intra-day schedule of the WDS; instead it relies on the third pump affinity law to adjust pump power upon activation of the balancing service. However, this relationship is approximate and prone to significant errors [22,23].

Finally, in [24], the authors combine both I/C-L and balancing services, with the study extended to desalination plants in [25]. However, these works are limited to the availability stage of the balancing service, neglecting its actual activation. In turn, this could result in violations of the tank operating limits during the activation stage.

1.3. Literature review: WDS hydraulic modeling

Beyond the above points regarding the accommodation of multiple grid services, the primary limitation in the existing literature concerns the hydraulic modeling of WDSs, specifically the assumption of full parameter availability. As later discussed in detail in Section 2.2, state-of-the-art works [8–14,16–21,24,25] presuppose accurate knowledge of the WDS parameters, particularly regarding pumps and pipelines. Nevertheless, in practice, this assumption is difficult to justify due to the inherent uncertainties of WDSs, which are further exacerbated by phenomena such as pipeline leakage and aging. Furthermore, this assumption leads to less accurate WDS modeling, while also increasing the complexity of the associated optimization problems for WDS scheduling and grid services provision.

A different approach is adopted in [15]. Specifically, the authors attempt to implicitly handle WDS uncertainties related to pump modeling by proposing a robust optimization formulation. Nonetheless, robust formulations result in more conservative solutions compared to their deterministic counterparts [26].

Table 1
Comparison between the existing literature and the proposed methodology.

Studies	Demand-side flexibility schemes			WDS modeling				
	Dynamic pricing	I/C-L service	Balancing service			VSPs	Pump shutdown	Measurement-based approach
			Day-ahead scheduling	Intra-day scheduling	Distinct upward & downward capacities			
[8–10]	✓	✗	✗	✗	✗	✗	✓	✗
[11–14], [15]	✓	✗	✗	✗	✗	✓	✓	✗
[16–18]	✓	✓	✗	✗	✗	✗	✓	✗
[19]	✓	✓	✗	✗	✗	✓	✓	✗
[20]	✓	✗	✗	✓	✗	✓	✗	✗
[21]	✓	✗	✓	✗	✗	✓	✓	✗
[24,25]	✗	✓	✓	✗	✓	✓	✗	✗
Proposed	✓	✓	✓	✓	✓	✓	✓	✓

1.4. Scope and contribution

The scope of this study is to enable WDSs to participate in both implicit and explicit demand-side flexibility schemes and to provide multiple grid services. To this end, the study addresses the identified gaps in both hydraulic modeling and scheduling of WDSs, presenting a two-fold contribution:

- **Measurement-based decoupling of VSP quantities and problem simplification:** To address the limitation of requiring full parameter knowledge for WDSs, this paper proposes a new measurement-based modeling approach. Specifically, the paper demonstrates that the electric power consumption and the flow rate of VSPs can be directly and accurately correlated through a convex quadratic function. This function decouples the flow rate and electric power consumption of VSPs from other quantities, such as rotational speed and hydraulic head. In turn, this decoupling significantly reduces the number of decision variables, thereby simplifying the formulation of the optimization problems for WDS scheduling. Consequently, the linearizations and relaxations introduced in the literature [8–14,16–21,24,25], which result from the inherent complexity of WDSs, can be avoided, allowing for exact solutions. Furthermore, it is highlighted that the proposed modeling approach aligns with the research guidelines of the IEEE PES Taskforce on Water-Power Systems [27,28], which emphasize the development of simple yet accurate models for the integration and optimization of WDSs and EPSs.
- **Two-stage scheduling for multiple services:** Contrary to the existing literature, the paper proposes a comprehensive framework that optimizes the operating schedule of WDSs by jointly considering all three common demand-side flexibility schemes, i.e., (a) dynamic electricity pricing, (b) I/C-L provision, and (c) balancing service provision. Furthermore, unlike previous studies focusing on balancing services [20,21,24,25], the proposed framework optimizes the WDS schedule for both the availability and the activation stages of the balancing service. To this end, it is structured as a two-stage problem. The first stage concerns the derivation of the day-ahead schedule, aiming to (a) minimize the net operating cost of the WDS, (b) provide I/C-L service, and reserve balancing capacity, and (c) ensure that tank operating limits are respected during the activation of the balancing service. The second stage is related to the intra-day activation of the balancing service upon its request from the EPS operator. The intra-day schedule ensures (a) the optimal allocation of pump setpoints for the balancing service activation and (b) the cost-effective adjustment of the WDS schedule for the remainder of the day. Moreover, contrary to [20,24,25], the proposed formulation accounts for the possibility of pump shutdown, thus enabling increased flexibility. Finally, leveraging the proposed measurement-based modeling approach,

the scheduling problems are formulated as mixed-integer quadratically constrained programming (MIQCP) optimization problems with convex continuous relaxations, facilitating computational tractability.

As an additional strength of the study, it is emphasized that both the proposed measurement-based approach for WDS modeling and the proposed optimization framework for WDS scheduling are applied and validated on a real-world WDS in Greece, as presented in Section 4.

Finally, to provide a clear comparison with the existing literature, Table 1 summarizes the contributions of the proposed methodology and the research gaps it addresses.

The remainder of the manuscript is organized as follows: Section 2 briefly reviews current practices in the modeling of WDSs and presents the proposed measurement-based modeling approach. Section 3 introduces the proposed two-stage optimization framework for WDS scheduling. In Section 4, the proposed methodology, including both the modeling approach and the optimization framework, is applied to and validated on a real-world WDS. Finally, Section 5 concludes the paper, summarizing the main findings. In addition, the application of the proposed methodology to a benchmark WDS is presented in Appendix A.

2. WDS hydraulic modeling

2.1. Preliminaries

In typical municipal WDSs [2,19,29], pumps in pumping stations extract water from supply sources (e.g., underground wells or reservoirs) and transfer it to storage tanks located at higher elevations. In turn, storage tanks define pressure zones [30] and supply consumers by gravity. When additional hydraulic head is required to transfer water to the tanks, booster pumps are employed.

In this configuration, the operation of a pump is defined by the hydraulic head (H) it generates and the water flow rate (Q), hereafter referred to as flow, that it delivers. In the case of VSPs, the motors are equipped with variable-frequency drives on their electric supply side. Thus, the rotational speed (n) of VSPs can be controlled in proportion to the input frequency of the drive, thereby allowing greater flexibility in terms of the generated head and the delivered flow.

More specifically, for VSPs, H , Q , and n are related via the pump affinity laws and the pump characteristic curve. The pump characteristic curve is commonly approximated by a bivariate quadratic polynomial, as shown in (1) [11], where n_r denotes the rated rotational speed. In (1), c_1 , c_2 , and c_3 are the coefficients of the bivariate quadratic polynomial and are specific to each pump.

$$H = c_1 \cdot Q^2 + c_2 \cdot \frac{n}{n_r} \cdot Q + c_3 \cdot \left(\frac{n}{n_r}\right)^2. \quad (1)$$

The pump efficiency (η) depends on both flow and rotational speed and can be approximated by (2), where e_1 , e_2 , and e_3 are coefficients specific

to each pump [31]:

$$\eta = e_1 \cdot \left(\frac{n_r}{n}\right)^3 \cdot Q^3 + e_2 \cdot \left(\frac{n_r}{n}\right)^2 \cdot Q^2 + e_3 \cdot \left(\frac{n_r}{n}\right) \cdot Q. \quad (2)$$

In practice, the hydraulic head that a VSP needs to generate in order to transfer water at flow Q through a pipeline to a tank at higher elevation can be approximated by the Darcy–Weisbach formula as

$$H = H_o + K \cdot Q^2. \quad (3)$$

Eq. (3) is commonly referred to as the system curve. The term $K \cdot Q^2$ represents head losses along the pipeline due to friction. The coefficient K is intrinsic to the pipeline and depends on its length, cross-sectional area, and material. Moreover, H_o represents the static head, that is, the total elevation difference between the suction side and the tank inlet. It is noted that, in WDSs, the tank inlet is typically positioned at the top of the tank, above the water level [9,20].

Consequently, if a VSP with characteristic curve (1) operates at speed n and transfers water through a pipeline with system curve (3), the operating point (H, Q) is defined by the intersection of curves (1) and (3). Finally, for a motor with efficiency η_m , the motor-pump system consumes electric power P :

$$P = \frac{\gamma \cdot H \cdot Q}{\eta \cdot \eta_m}, \quad (4)$$

with γ denoting the specific weight of water.

2.2. Current practices

To model WDSs, existing studies [8–14,16–21,24,25] simultaneously employ (1), (3), and (4), for all pumps and pipelines. Nonetheless, (2) is omitted and a constant efficiency η is instead assumed to simplify the model. Likewise, a constant η_m value is assumed for the motor. This conventional modeling approach exhibits the following drawbacks:

- **Dependence on parameter availability:** It heavily relies on the availability and accurate knowledge of parameters c_1 , c_2 , c_3 , K , and H_o , for all pumps and pipelines. Nonetheless, this assumption is difficult to justify for several reasons. First, WDSs are known to suffer significant leakage, with average water losses of around 30% [32]. The presence of leaks distorts pipeline characteristics, leading to significant deviations from the theoretical values. In addition to leakages, WDS models, if available, require frequent calibration and parameter updates [33], to account for discrepancies due to pipeline aging and damage. Moreover, the static head parameter is subject to uncertainty [34], with errors of a few meters potentially causing significant inaccuracies in the calculations of WDS quantities [35]. Finally, although pump curves are usually provided by manufacturers, they have been identified as sources of uncertainty [15,36].
- **Complex problem formulation:** In addition to the assumption of accurate knowledge of such an extensive set of uncertain WDS parameters, the conventional modeling approaches introduce a substantial number of variables into the optimization problem of WDS scheduling. More specifically, for each pump, Q , H , P , and n are decision variables to be calculated, while η is incorrectly assumed to be constant. The increased number of variables, along with the significant nonlinearity of (1), (3), and (4), renders such problems difficult to solve. For this reason, existing studies resort to extensive approximations and relaxations to the original problem, in order to achieve computationally tractable solutions. As a trade-off, such approximations result in errors and discrepancies with respect to the actual physical model of the WDS.
- **Neglect of efficiency reduction:** Pump efficiency decreases at reduced speeds [23], with the reduction being more significant in systems with substantial static head [37]. Thus, assuming a constant η leads to important errors in power calculations. Motor efficiency η_m also decreases at reduced speeds due to the reduction in power, albeit less significantly.

2.3. Proposed modeling approach

To avoid reliance on uncertain parameters, this paper proposes a new measurement-based modeling approach for WDS, deriving a simplified yet equivalent model. The proposed modeling approach is based on the principles stipulated in [38], which are also leveraged in [39,40]. Specifically, the aim is to decouple the interactions of interconnected pumps, pipelines, and storage tanks by implicitly representing the hydraulic behavior of the WDS within an equivalent pump operating model. The proposed modeling approach can be applied to WDSs with the typical configuration described in Section 2.1, which has been highlighted for its potential in demand-side flexibility [1].

To elaborate, an on-site, measurement-based, system-learning procedure is developed, as shown in Fig. 1(a). For each pump-pipeline pair in the WDS, this procedure derives a dedicated $P = g(Q)$ function that correlates the flow delivered by the pump with the corresponding electric power consumed by its motor. Specifically, commencing from the nominal value (f_n), the frequency applied to the VSP is iteratively reduced in steps of Δf . For each input frequency, the corresponding flow and electric power operating points are recorded and stored in a lookup table. The process is repeated until either the pump operating flow or frequency is reduced below the rated minimum value ($Q_{\min,r}$ and $f_{\min,r}$, respectively), as specified by the manufacturer. The step size Δf can be set to 1 Hz or lower to ensure high resolution and accurate modeling. The derived $P = g(Q)$ curve is fixed for a specific pump-pipeline pair. Therefore, the curves need to be extracted only once and updated according to the recommended practices for WDS model calibration, typically once every one to five years [41], to account for pipeline aging and damage. Additionally, since modern WDSs are equipped with Supervisory Control and Data Acquisition systems, non-scheduled calibrations can be performed whenever the flow values stored in the lookup tables do not match the measured values, indicating model discrepancies. The proposed learning process:

- achieves simpler yet more accurate WDS modeling, as it relies on up-to-date measurements and implicitly accounts for the decrease in pump and motor efficiencies at reduced speeds.
- decouples decision variables P and Q from H , n , and η , implicitly accounting for their interdependence. As a result, the number of variables involved in the optimization problems is reduced by 60% per pump.

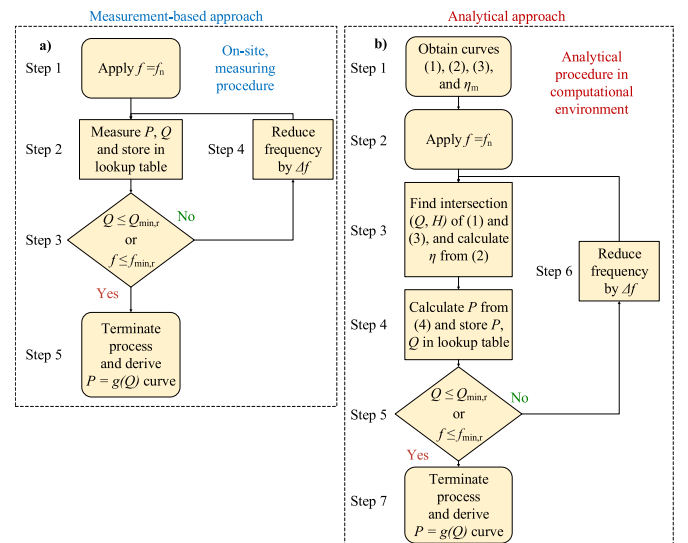


Fig. 1. a) Proposed measurement-based approach and b) analytical alternative.

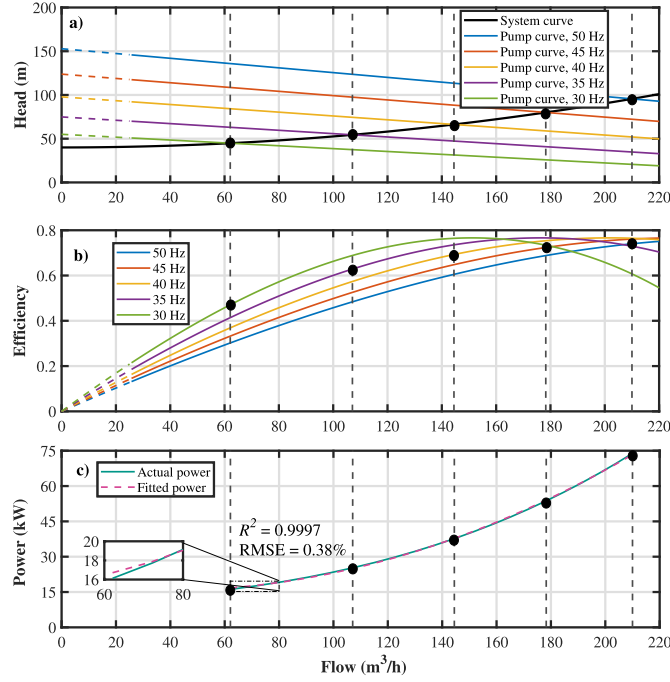


Fig. 2. Extraction of $P = g(Q)$ curve for a typical pump-pipeline pair using the analytical approach of Fig. 1(b).

Alternatively, even if the assumption of full parameter availability holds, such a procedure for the extraction of $P = g(Q)$ curves should be performed, in order to simplify the formulation of the optimization problem for WDS scheduling. More specifically, under complete parameter availability, the measurement-based approach of Fig. 1(a) can be replaced by the analytical computational approach of Fig. 1(b). In this case, the $P = g(Q)$ curve is mathematically derived, leveraging the knowledge of (1), (2), and (3).

An illustrative example of the analytical approach of Fig. 1(b) is presented in Fig. 2 for a typical pump-pipeline pair. Therefore, the example of Fig. 2 is not based on measurements but rather on analytical pump characteristic curves, pump efficiency curves, and system curves, to demonstrate the theoretical interrelation among the physical quantities H , Q , η , n , and P . Specifically, Fig. 2(a) illustrates the characteristic curves (1) of a typical water pump (Grundfos SP 270-G-3A), at five distinct input frequency values. In addition, a typical system curve (3) is superimposed. For each input frequency, the pump operating point (H , Q), marked by a black dot, is determined by the intersection of the corresponding pump curve with the system curve.

Regarding pump efficiency, Fig. 2(b) presents the efficiency curves (2) for the same five input frequency values. For each pump operating point of Fig. 2(a), the pump efficiency is identified by the point on the respective curve of Fig. 2(b) at the Q value of the operating point. It is observed that pump efficiency decreases significantly with reduced frequency. In this case, efficiency dropped from 0.74 at 50 Hz to 0.46 at 30 Hz, i.e., by 38%. As discussed, such reductions in efficiency are not taken into account by existing studies, since they assume a constant pump efficiency.

Finally, the intersection points between the pump curves and the system curve, along with the calculation of the corresponding efficiency values, enable the derivation of the $P = g(Q)$ curve, as demonstrated in Fig. 2(c). Furthermore, the flow and electric power profiles with respect to frequency, as stored in the lookup table, are presented in Fig. 3. In general, the $P = g(Q)$ curve can be accurately approximated by a

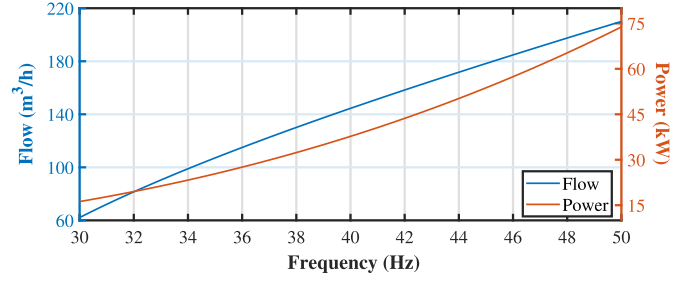


Fig. 3. Flow and power versus frequency, as stored in the lookup table.

second-order polynomial, as follows:

$$P = g(Q) = a \cdot Q^2 + b \cdot Q + c. \quad (5)$$

For the representative case of Fig. 2, the second-order polynomial fit is also plotted in Fig. 2(c). As demonstrated, the model achieves a very accurate fit, as verified by the coefficient of determination ($R^2 \simeq 1$), and the root-mean-square error (RMSE = 0.38%). The RMSE is normalized with respect to the maximum operating power value at 50 Hz. Similar fitting accuracies are observed for different pump-pipeline pairs, as well as for the real-world cases in Section 4. Moreover, it is noted that higher-order polynomials can achieve better fitting accuracy for the $P = g(Q)$ curve due to their greater degrees of freedom, although the improvement is marginal. Thus, a second-order polynomial fit is selected, as it ensures both high accuracy, as shown in Fig. 2(c), and modeling simplicity, as discussed in Section 3.

Furthermore, as illustrated in Fig. 1, the $P = g(Q)$ extraction process terminates either when the minimum pump flow or frequency is reached. The termination point defines the actual operating minimum $[Q^{\min}, f^{\min}, P^{\min}]$, with $Q^{\min} \geq Q_{\min,r}$ and $f^{\min} \geq f_{\min,r}$. Moreover, pumps can be switched off; thus the complete operating range is discontinuous, defined as

$$Q^k \in \{0\} \cup [Q^{\min}, Q^{\max}]; \quad P^k \in \{0\} \cup [P^{\min}, P^{\max}]. \quad (6)$$

In (6), Q^k and P^k denote the flow and power, respectively, including the case of pump shutdown. $Q^{\max}$ and $P^{\max}$ are the corresponding operating points at $f = f_n$.

Finally, it is noted that the proposed measurement-based modeling approach can be applied to pumps operating in parallel and delivering flow through a common pipeline. In this case, an equivalent $P = g(Q)$ curve can be derived to represent the aggregate operation of the pump group and the common pipeline.

3. Proposed optimization framework for WDS scheduling

The proposed optimization framework schedules WDSs to enable their participation in both implicit and explicit demand-side flexibility schemes across distinct time horizons, incorporating the proposed modeling approach presented in Section 2.3. The framework formulates day-ahead and intra-day scheduling problems, which are solved from the perspective of the WDS operator to minimize the net operating cost of the WDS.

3.1. Examined demand-side flexibility schemes

As discussed in Section 1.4, the proposed optimization framework accommodates three demand-side flexibility schemes: (a) dynamic electricity pricing, (b) I/C-L service provision, and (c) balancing service provision.

As explained in Section 1.1, dynamic electricity pricing is an implicit demand-side flexibility scheme. In this scheme, the WDS is encouraged

to adjust its energy consumption based on the time variation of electricity tariffs, thereby implicitly contributing to the EPS operation, e.g., through load profile flattening.

In contrast, the I/C-L and balancing services are explicit demand-side flexibility schemes. To enhance the generality of the proposed framework, different structures are considered for these two grid services. Specifically, the I/C-L service involves an agreement for the WDS to reduce its energy consumption during specified, predetermined time intervals, in line with schemes such as [42]. These specified time intervals are known in advance, i.e., in the day-ahead stage, as they typically correspond to periods with high load demand for the EPS. The available power reduction is calculated in the day-ahead stage and delivered in the intra-day stage.

On the other hand, the balancing service is structured as a ramping or frequency restoration reserve product [43], representing the ability of the WDS to provide upward and downward reserve capacity by decreasing or increasing its power consumption, respectively, in accordance with the generator convention. Specifically, the balancing service involves the day-ahead reservation of balancing capacity, and the potential intra-day activation of this capacity at unknown time intervals, if requested by the EPS operator. As a result, the activation is stochastic, addressing unforeseen conditions in the EPS that cannot be known in advance.

Thus, the optimization framework accommodates the joint provision of grid services with both deterministic (I/C-L service) and stochastic (balancing service) deliveries in the intra-day stage. The differences in the purpose and structure of each service are further analyzed in Section 3.2.

3.2. Day-ahead stage

The day-ahead optimization determines the WDS operating schedule for the following day, aiming to minimize the net daily operating cost. The actual operating cost refers to the energy consumption of the pumps, which is billed under a dynamic pricing scheme. In addition, two revenue sources are considered, corresponding to the provision of I/C-L and balancing services.

Let $\mathcal{N}_p$ and $\mathcal{N}_\tau$ denote the sets of pumps and tanks of the WDS, respectively. Moreover, for each tank $\tau \in \mathcal{N}_\tau$, the sets $\mathcal{N}_{in,\tau} \subseteq \mathcal{N}_p$ and $\mathcal{N}_{out,\tau} \subseteq \mathcal{N}_p$ represent the pumps at the inlet (upstream) and outlet (downstream) of the tank, respectively. The set $\mathcal{T}$ denotes the daily time horizon. The set $\mathcal{T}_{I/C-L} \subseteq \mathcal{T}$ denotes the specified time period for the provision of the I/C-L service, and the set $\mathcal{T}_R \subseteq \mathcal{T}$ denotes the time period for the provision of the balancing service. Typically, $\mathcal{T}_{I/C-L}$ corresponds to periods of high load for the EPS, although its exact definition varies across different I/C-L schemes. Furthermore, it is assumed that the two services do not overlap; thus $\mathcal{T}_{I/C-L} \cap \mathcal{T}_R = \emptyset$ and $\mathcal{T}_R = \mathcal{T} - \mathcal{T}_{I/C-L}$.

For a pump $p \in \mathcal{N}_p$, the flow and the electric power consumption are represented by two sets of variables. In line with (6), continuous variables $Q_p[t]$, $P_p[t]$ model the operation of the pump under different input frequencies, while mixed discrete-continuous variables $Q_p^k[t]$, $P_p^k[t]$ are used to include the possibility of shutdown. More specifically, pump shutdown is modeled via binary variables $k_p[t]$.

For a tank $\tau \in \mathcal{N}_\tau$, the decision variable is the normalized volume of stored water relative to the total tank volume V_τ , expressed in per unit (p.u.). This normalized volume is referred to as the state-of-charge (SoC) and is denoted as $SoC_\tau[t]$, by analogy with energy storage systems, since tanks can also be considered storage modules for gravitational potential energy.

Finally, at the WDS level, the provision of the I/C-L service is quantified via the decision variable $P_{I/C-L}[t]$, which physically represents the capacity of the WDS to reduce its power consumption during the designated $\mathcal{T}_{I/C-L}$ time period. Regarding the balancing service, it is quantified via the total upward and downward reserve capacities of the WDS, denoted by the decision variables $R_u[t]$ and $R_d[t]$, respectively. These capacities physically represent the ability of the WDS to decrease

or increase its power consumption at $t \in \mathcal{T}_R$, in response to a request from the EPS operator.

The objective function of the day-ahead problem is:

$$\min \sum_{t \in \mathcal{T}} \left(\sum_{p \in \mathcal{N}_p} P_p^k[t] \right) \cdot C[t] \cdot \Delta t - \sum_{t \in \mathcal{T}_{I/C-L}} P_{I/C-L}[t] \cdot C_{I/C-L}[t] \cdot \Delta t - \sum_{t \in \mathcal{T}_R} (R_u[t] \cdot C_u[t] + R_d[t] \cdot C_d[t]) \cdot \Delta t. \quad (7)$$

In (7), the first term represents the energy cost of pumping, while the second and third terms are the economic benefits from the I/C-L and balancing services, respectively. Moreover, the parameter $C[t]$ is the time-varying electricity price. For the I/C-L and balancing services, the power reduction and the upward and downward reserve capacities are compensated at rates $C_{I/C-L}[t]$, $C_u[t]$, and $C_d[t]$, respectively. Finally, Δt denotes the analysis time step. In line with common practices for balancing services, Δt should be sub-hourly, typically between 5 and 15 minutes. After the day-ahead problem is solved, the derived quantities $P_{I/C-L}[t]$, $R_u[t]$, and $R_d[t]$ are nominated to the EPS operator.

As discussed in Section 3.1, in the I/C-L service, the WDS is remunerated for reducing its consumption during specified time intervals. I/C-L services [42] define this incentivized power reduction with respect to a reference, baseline power consumption, as it is also practiced in relevant literature [18,19]. This baseline power is generally calculated based on historical data, with its definition varying across different I/C-L schemes. For the WDS, this baseline power consumption is denoted as $P_b[t]$, while the power reduction capacity of the I/C-L service is defined in (8). Regarding $P_{I/C-L}[t]$, it is considered constant $\forall t \in \mathcal{T}_{I/C-L}$ to achieve a reduction in the WDS consumption with a stable rate.

$$P_{I/C-L}[t] = P_b[t] - \sum_{p \in \mathcal{N}_p} P_p^k[t], \quad t \in \mathcal{T}_{I/C-L}. \quad (8)$$

with $P_{I/C-L}[t] \text{ constant } \forall t \in \mathcal{T}_{I/C-L}$.

The upward and downward reserve capacities $R_u[t]$ and $R_d[t]$ are defined in (9). As observed, in contrast to $P_{I/C-L}[t]$, they are not defined with respect to a baseline. Instead, they represent the operating foot-room and headroom of the WDS to decrease or increase its power consumption at $t \in \mathcal{T}_R$, as the service addresses the need for available balancing power during unforeseen conditions in the EPS.

$$R_u[t] = \sum_{p \in \mathcal{N}_p} P_p^k[t]; \quad R_d[t] = \sum_{p \in \mathcal{N}_p} (P_p^{\max} - P_p^k[t]). \quad (9)$$

Main constraints: Eqs. (10)–(13) formulate the technical constraints of the WDS. More specifically, (10) represents the time evolution of the tank SoC. Parameter $Q_\tau^c[t]$ denotes the day-ahead aggregate forecast water demand to be supplied by tank τ , while parameter SoC_τ^0 denotes the initial SoC value of the tank, corresponding to the final value from the previous day. The second line of (10) enforces the recovery of the SoC. More specifically, t_f and t_0 are the final and initial intervals of $\mathcal{T}$, respectively. Enforcing SoC recovery to the initial value of the day prevents its reduction to the minimum operating limit. This constraint ensures sufficient headroom at the onset of each daily horizon for both increasing and decreasing the WDS power consumption.

$$SoC_\tau[t + \Delta t] = SoC_\tau[t] + \frac{\Delta t}{V_\tau} \left(\sum_{p \in \mathcal{N}_{in,\tau}} Q_p[t] - Q_\tau^c[t] - \sum_{p \in \mathcal{N}_{out,\tau}} Q_p[t] \right), \\ SoC_\tau[t_f] = SoC_\tau[t_0] = SoC_\tau^0, \quad \tau \in \mathcal{N}_\tau, \quad t \in \mathcal{T}. \quad (10)$$

As analyzed in Section 2.3, the flow delivered by a pump and the electric power consumed by its motor are correlated through a unique quadratic polynomial function, as in (11).

$$P_p[t] = a_p \cdot Q_p^2[t] + b_p \cdot Q_p[t] + c_p, \quad p \in \mathcal{N}_p, \quad t \in \mathcal{T}. \quad (11)$$

To model pump shutdown and avoid bilinear multiplications among decision variables $k_p[t]$, $P_p[t]$, and $Q_p[t]$, the equivalent, linearized constraints of (12) are used for $p \in \mathcal{N}_p$, $t \in \mathcal{T}$, as in [19]. Constraints (12)

equivalently set $P_p^k[t]$ equal to $P_p[t]$ or zero, when $k_p[t] = 1$ or $k_p[t] = 0$, respectively.

$$P_p^k[t] \leq P_p[t] - P_p^{\min} \cdot (1 - k_p[t]), \quad (12a)$$

$$P_p^k[t] \geq P_p[t] - P_p^{\max} \cdot (1 - k_p[t]), \quad (12b)$$

$$P_p^k[t] \leq P_p^{\max} \cdot k_p[t], \quad (12c)$$

$$P_p^k[t] \geq P_p^{\min} \cdot k_p[t], \quad (12d)$$

$$Q_p^k[t] \leq Q_p[t] - Q_p^{\min} \cdot (1 - k_p[t]), \quad (12e)$$

$$Q_p^k[t] \geq Q_p[t] - Q_p^{\max} \cdot (1 - k_p[t]), \quad (12f)$$

$$Q_p^k[t] \leq Q_p^{\max} \cdot k_p[t], \quad (12g)$$

$$Q_p^k[t] \geq Q_p^{\min} \cdot k_p[t]. \quad (12h)$$

Finally, constraints (13) impose the operating ranges for the SoC of tanks, as well as for the electric power consumption and flow of pumps. For pumps, the actual operating limits of (6), as derived from the proposed process of Fig. 1, are considered, rather than the rated values indicated by the manufacturer. For tanks, the operating range is decided by the WDS operator.

$$SoC_{\tau}^{\min} \leq SoC_{\tau}[t] \leq SoC_{\tau}^{\max}, \quad \tau \in \mathcal{N}_{\tau}, t \in \mathcal{T} \quad (13a)$$

$$P_p^{\min} \leq P_p[t] \leq P_p^{\max}, \quad p \in \mathcal{N}_p, t \in \mathcal{T} \quad (13b)$$

$$Q_p^{\min} \leq Q_p[t] \leq Q_p^{\max}, \quad p \in \mathcal{N}_p, t \in \mathcal{T} \quad (13c)$$

In summary, constraints (10) and (13a) ensure that the water demand is met while the operating limits of the tank SoC are respected. Furthermore, constraints (11)–(12) and (13b)–(13c) enforce the pump physical laws via the proposed modeling approach of Section 2.3.

Reserve activation constraints: Eq. (9) refers only to the reservation of capacities $R_u[t]$ and $R_d[t]$, as planned the day before. Subsequently, during the intra-day stage, the EPS operator may request the activation of the balancing service at a $t \in \mathcal{T}_R$, which is unknown during the intra-day stage. Therefore, additional constraints must be introduced into the day-ahead problem, to ensure that the nominated capacities $R_u[t]$ and $R_d[t]$ will not result in violations of the tank SoC limits upon reserve activation at any $t \in \mathcal{T}_R$. According to (9), if the nominated capacity $R_u[t]$ is requested, pumps must be shut down. In contrast, if the nominated capacity $R_d[t]$ is requested, pumps must be set at their maximum operating point, i.e., at $P_p^{\max}$, $Q_p^{\max}$. To this end, (14) recalculates the tank SoC for upward activation with $R_u[t]$ and downward activation with $R_d[t]$, for all $\tau \in \mathcal{N}_{\tau}$ and at any $t \in \mathcal{T}_R$. The recalculated SoC values are denoted as $SoC_{\tau}^u[t]$ and $SoC_{\tau}^d[t]$, for upward and downward activation, respectively. Finally, by forcing these recalculated SoC values to satisfy (15), the WDS schedule enables the provision of reserve capacity while respecting the tank operating limits.

$$SoC_{\tau}^u[t + \Delta t] = SoC_{\tau}[t] - \frac{\Delta t}{V_{\tau}} Q_{\tau}^c[t],$$

$$SoC_{\tau}^d[t + \Delta t] = SoC_{\tau}[t] + \frac{\Delta t}{V_{\tau}} \left(\sum_{p \in \mathcal{N}_{in,\tau}} Q_p^{\max} - Q_{\tau}^c[t] - \sum_{p \in \mathcal{N}_{out,\tau}} Q_p^{\max} \right), \quad (14)$$

$$SoC_{\tau}^u[t_0] = SoC_{\tau}^d[t_0] = SoC_{\tau}^0, \quad \tau \in \mathcal{N}_{\tau}, t \in \mathcal{T}_R.$$

$$SoC_{\tau}^{\min} \leq SoC_{\tau}^u[t] \leq SoC_{\tau}^{\max}, \quad \tau \in \mathcal{N}_{\tau}, t \in \mathcal{T}_R. \quad (15)$$

$$SoC_{\tau}^{\min} \leq SoC_{\tau}^d[t] \leq SoC_{\tau}^{\max},$$

Convexification: As illustrated in Fig. 2(c), the quadratic functions $P = g(Q)$ are convex, with $a_p \geq 0$ for any $p \in \mathcal{N}_p$. Nevertheless, for the feasible set of (11) to be convexified, the equality constraint must be relaxed as follows:

$$P_p[t] \geq a_p \cdot Q_p^2[t] + b_p \cdot Q_p[t] + c_p, \quad p \in \mathcal{N}_p, t \in \mathcal{T}. \quad (16)$$

To replace (11) with (16) in the problem formulation and preserve equivalence between the original problem and its convex reformulation, the

solution to the convex problem must satisfy (16) exactly with equality. For almost all relevant cases, this condition is inherently satisfied due to the structure of the objective function. Specifically, by substituting the expressions from (8)–(9) and because $\mathcal{T} = \mathcal{T}_{I/C-L} \cup \mathcal{T}_R$, the objective function (7) can be expanded as:

$$\begin{aligned} \min \quad & \sum_{t \in \mathcal{T}_{I/C-L}} \left(\sum_{p \in \mathcal{N}_p} P_p^k[t] \right) \cdot (C[t] + C_{I/C-L}[t]) \cdot \Delta t \\ & + \sum_{t \in \mathcal{T}_R} \left(\sum_{p \in \mathcal{N}_p} P_p^k[t] \right) \cdot (C[t] + C_d[t] - C_u[t]) \cdot \Delta t \\ & - \sum_{t \in \mathcal{T}_{I/C-L}} P_b[t] \cdot C_{I/C-L}[t] \cdot \Delta t - \sum_{t \in \mathcal{T}_R} \left(\sum_{p \in \mathcal{N}_p} P_p^{\max} \right) \cdot C_d[t] \cdot \Delta t. \end{aligned} \quad (17)$$

Therefore, as shown in (17), the factor $C[t] + C_{I/C-L}[t]$ penalizes $P_p^k[t]$, $\forall t \in \mathcal{T}_{I/C-L}$, thus driving $P_p[t]$ to its lower quadratic bound. For $\forall t \in \mathcal{T}_R$, the factor $C[t] + C_d[t] - C_u[t]$ is typically positive for most market scenarios, as demonstrated by several EPS datasets [44,45]. Consequently, the objective function drives $P_p[t]$ to its lower quadratic bound $\forall t \in \mathcal{T}_R$ as well, allowing (11) to be equivalently relaxed as (16) for most relevant market scenarios.

In conclusion, the proposed day-ahead problem:

$$\begin{array}{ll} \text{Proposed} & \min \quad (7) \\ \text{day-ahead problem} & \text{s.t.} \quad (8)–(10), \quad (12)–(16). \end{array}$$

is an MIQCP problem with convex continuous relaxations, facilitating computational tractability. Equivalently, it can be formulated as a mixed-integer second-order cone program.

3.3. Intra-day stage

The intra-day stage of the proposed optimization framework refers to a time horizon $\mathcal{T}_{id} \subseteq \mathcal{T}$, which commences at the time interval $t_a \in \mathcal{T}_R$ when the EPS operator requests the activation of the balancing service. The intra-day problem aims to optimize the pump setpoints for the requested reserve activation and to adjust the pump schedule for the remainder of the day after reserve activation, to: (a) minimize the operating cost; (b) ensure SoC recovery to its initial value at the end of the day; (c) ensure the delivery of the I/C-L service, if t_a occurs before $\mathcal{T}_{I/C-L}$. The activation of the balancing service is represented by (18a) and (18b) for the upward and downward cases, respectively. The parameter R^a denotes the balancing power requested by the EPS operator. The parameter $P_{s,a}^{DA}$ is the total WDS power consumption at t_a according to the day-ahead schedule, while $P_s^{\min}$ is equal to $\sum_{p \in \mathcal{N}_p} P_p^{\min}$. The feasible range of R^a for upward activation is discontinuous because the pump operating range is also discontinuous.

$$\sum_{p \in \mathcal{N}_p} P_p^k[t_a] = P_{s,a}^{DA} - R^a, \quad R^a \in [0, R_u[t_a] - P_s^{\min}] \cup \{R_u[t_a]\} \quad (18a)$$

$$\sum_{p \in \mathcal{N}_p} P_p^k[t_a] = P_{s,a}^{DA} + R^a, \quad 0 \leq R^a \leq R_d[t_a] \quad (18b)$$

Moreover, for the intra-day problem, the SoC time evolution for tank τ is represented by (19), with SoC_{τ}^a being the SoC value just prior to the onset of the activation interval.

$$SoC_{\tau}[t + \Delta t] = SoC_{\tau}[t] + \frac{\Delta t}{V_{\tau}} \left(\sum_{p \in \mathcal{N}_{in,\tau}} Q_p[t] - Q_{\tau}^c[t] - \sum_{p \in \mathcal{N}_{out,\tau}} Q_p[t] \right), \quad (19)$$

$$SoC_{\tau}[t_a] = SoC_{\tau}^a, \quad \tau \in \mathcal{N}_{\tau}, t \in \mathcal{T}_{id}.$$

Furthermore, the problem includes (12)–(13), and (16), for $t \in \mathcal{T}_{id}$. Additionally, if the reserve activation request occurs before the onset

of the $\mathcal{T}_{I/C-L}$ period, constraint (8) must also be included in the intra-day problem, but with $P_{I/C-L}[t]$ as a known parameter determined from the day-ahead stage. The inclusion of constraint (8) ensures the delivery of the I/C-L service. Finally, as observed, SoC recovery is not explicitly enforced in (19), as infeasibilities may arise in case of late reserve activation. Thus, similarly to [21], a penalty term with weight c_τ is included in the objective function to indirectly enforce SoC recovery:

$$\min \sum_{t \in \mathcal{T}} \left(\sum_{p \in \mathcal{N}_p} P_p^k[t] \right) \cdot C[t] \cdot \Delta t + \sum_{\tau \in \mathcal{N}_\tau} c_\tau \cdot (SoC_\tau[t_f] - SoC_\tau^0)^2. \quad (20)$$

It is noted that (20) penalizes $P_p^k[t]$, thus driving $P_p[t]$ to its lower quadratic bound and enabling the use of the relaxed (16). Thus, the proposed intra-day problem:

$$\begin{aligned} \min \quad & (20) \\ \text{Proposed} \quad & \\ \text{intra-day problem} \quad & \text{s.t.} \quad (12)-(13), (16)-(19), \\ & \text{and (8) if } t_a \text{ before } \mathcal{T}_{I/C-L}. \end{aligned}$$

is an MIQCP problem with convex continuous relaxations.

3.4. Discussion on modeling

As discussed in Section 2.3 and demonstrated in Section 3.2 and Section 3.3, the proposed modeling approach decouples the physical quantities of pump hydraulic head and frequency from the day-ahead and intra-day optimization problems. To elaborate, the pump input frequency is implicitly represented by the quadratic $P = g(Q)$ curve and is calculated a posteriori for each operating point, using the lookup tables derived in Step 2 of Fig. 1(a).

Similarly, the hydraulic head generated by the pump is also implicitly represented by the $P = g(Q)$ curve. Specifically, each point on the curve represents a different valid operating point of the corresponding pump-pipeline pair, and thus a different valid and physically consistent operating head, as visualized in Fig. 2(a) and (c).

Regarding the entire WDS, as per the principles established in [30,38,39,46,47], for well-designed WDSs, any pump operating schedule that is feasible with respect to the tank water level constraints also satisfies the hydraulic head constraints at all junctions. This principle also applies to consumer junctions downstream of storage tanks. To elaborate, consumer water demands can be delivered at an appropriate pressure from the storage tanks even when pumps are not operating, since tanks are located at higher elevations [47]. Therefore, junctions downstream of tanks can be modeled as an aggregate water demand $Q_c^c[t]$ to be supplied by the tanks, since the downstream flow is gravity-fed and does not involve pump operation or electricity consumption.

As a result, the physically feasible $P = g(Q)$ curve ensures that pumps generate sufficient head to supply the tanks. In turn, SoC constraints (10), (14), and (19) ensure that tank water level limits are satisfied. Therefore, the proposed optimization framework implicitly ensures that hydraulic head requirements are satisfied throughout the WDS, in accordance with the principles of [30,38,39,46,47].

4. Results and validation

In this section, the proposed methodology, including both the measurement-based modeling approach and the two-stage optimization framework, is validated. All examined optimization problems are formulated in the General Algebraic Modeling System (GAMS) and solved using the SCIP solver on a personal computer with an AMD Ryzen 9 7940HS (8C/16T, 4.0–5.2 GHz) processor and 32 GB RAM.

4.1. Real-world system under study and input parameters

The proposed methodology is applied and validated on a real-world WDS, serving the town of Igoumenitsa, in northwestern Greece. The WDS comprises 12 pumps (11 groundwater extraction stations and 1

booster pump), 5 tanks, and 5 aggregate water consumption points, as illustrated in Fig. 4. As observed, the WDS consists of four independent subsystems, each pumping from a distinct underground source to supply a different part of the town. Despite the hydraulic independence of these subsystems, the proposed optimization framework couples them as flexible loads of the EPS via (8) and (9), enabling the system-wide calculation of capacities for I/C-L and balancing services.

The maximum operating flow and electric power of the pumps, as well as the volumes of the tanks, are presented in Fig. 4. For the water demands $Q_c^c[t]$, actual time series recorded by the Igoumenitsa WDS operator on 2 August 2024 are used, as presented in Fig. 5(a) and provided in [48]. For the electricity cost $C[t]$ and the reserve compensation rates $C_u[t]$ and $C_d[t]$, clearing prices from the day-ahead market of the California Independent System Operator for the same date are used, as shown in Fig. 5(b) [44]. As observed, the reserve compensation rates for this day are considerably lower than the corresponding electricity prices. The upward compensation rate increases significantly between 18:00 and 21:00, while downward rates are higher for the remaining hours. Moreover, the electricity price is also higher between 18:00 and 21:00. Therefore, in this study, the time period $\mathcal{T}_{I/C-L}$ of the I/C-L service is defined as 18:00–21:00, as higher electricity prices indicate a greater need for the I/C-L service. Furthermore, $C_{I/C-L}[t]$ is set to 200 \$/MWh, $\forall t \in \mathcal{T}_{I/C-L}$. Regarding the baseline power consumption $P_b[t]$, without loss of generality, it is assumed to be $0.5 \cdot \sum_{p \in \mathcal{N}_p} P_p^{\max}$, $\forall t \in \mathcal{T}_{I/C-L}$, representing the midpoint of the maximum WDS power consumption. As described in Section 4.3, this assumption is consistent with the current operating practice of the Igoumenitsa WDS, which involves switching pumps on and off based on the SoC level of the downstream tanks, thereby resulting in an average power consumption around this midpoint. Finally, the analysis time step Δt is set to 15 minutes.

For all tanks, the SoC limits are set to $SoC_\tau^{\max} = 0.9$ p.u. and $SoC_\tau^{\min} = 0.2$ p.u., to enable flexibility while preventing the risks of depletion and overflow [29]. The initial SoC level of tanks is assumed to be $SoC_\tau^0 = 0.5$ p.u.

4.2. Validation of the modeling approach

For all 12 pump-pipeline pairs, the proposed modeling approach is applied through on-site measurements with $\Delta f = 1$ Hz. Power consumption is measured using the Metrel MI 2892 Power Master power analyzer, which provides Class 1 metering accuracy. Flow is measured using the already installed electromagnetic meters of the Igoumenitsa

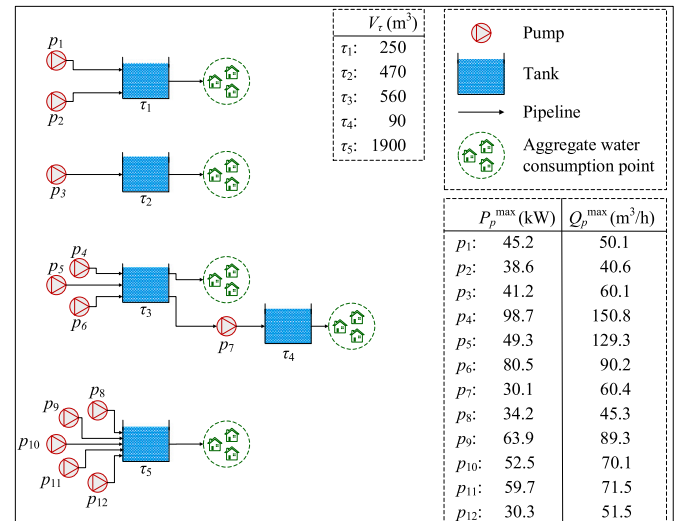


Fig. 4. Schematic of the real-world WDS under study.

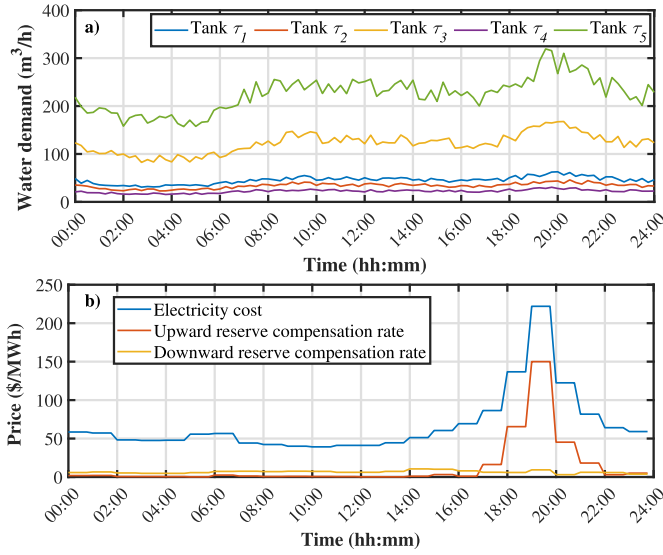


Fig. 5. (a) Aggregate water demand $Q_c^e[t]$ to be supplied by tank τ ; (b) prices $C[t]$, $C_u[t]$, and $C_d[t]$.

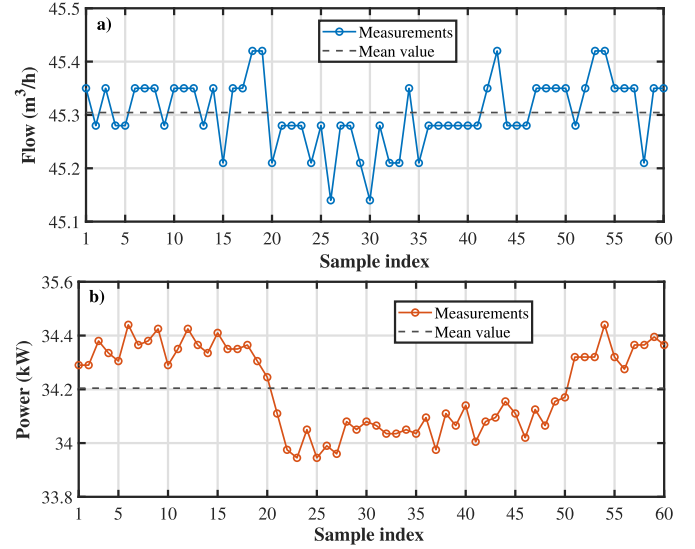


Fig. 6. (a) flow and (b) power measurements recorded every 5 s over 5 minutes for pump 8 at 50 Hz.

WDS, specifically the Siemens MAG5100 and MAG3100. The meters are coupled with the Siemens MAG6000 transmitter. The percentage error introduced by these meters is typically below 0.25%. For each applied input frequency, flow and power are recorded for 5 minutes at a sampling interval of 5 s, thus yielding 60 measurement points. An indicative example of the measuring process is illustrated in Fig. 6, which depicts the flow and power of Pump 8 for $f = 50$ Hz, measured over 5 minutes. As observed, the deviation of the measurements from their mean value is minimal, namely lower than 0.5%, due to the high precision of the meters involved. Thus, to derive the final (Q, P) operating point for each frequency and forward them to the lookup table of the pump, the mean value of the 60 measurement points is used.

As described in Section 2.3, the measurement-acquisition process terminates when either the minimum flow or input frequency value of the pump has been reached. Subsequently, the $P = g(Q)$ quadratic curves are derived. The quadratic coefficients and the minimum operating flow for each pump are listed in Table 2. For all pumps, the accuracy of the proposed modeling approach is validated by the quality of the quadratic fits, which is in turn confirmed by the high R^2 values and low RMSE values. Indicatively, the quadratic fit to the measurements of pumps p_2 and p_5 is presented in Fig. 7(a). Moreover, Fig. 7(b) illustrates the lookup tables that relate power and flow to the input frequency of the pumps. The accuracy metrics of the quadratic fit for all 12 pumps are presented in Table 3. The flow and electric power measurements of the 12 pumps can be found in [48].

4.3. Validation of the day-ahead stage

Here, the day-ahead scheduling stage is examined. To validate the effectiveness of the proposed optimization framework in enabling WDSs to provide multiple grid services, the schedules derived from the day-ahead problem of Section 3.2 are compared with two base cases.

Base Case 1 corresponds to the current operational practice for WDSs. Specifically, pumps are controlled via float switches based on the SoC of the downstream tanks. The SoC fluctuates within a permissible range, and the upstream pumps are automatically switched on or off when the SoC falls below or exceeds the lower and upper bounds, respectively. Base Case 1 is evaluated using two different sets of SoC bounds. In Base Case 1a, the actual bounds currently adopted by the Igoumenitsa WDS operator are used, namely, 0.4 p.u. and 0.6 p.u. As observed, due to the absence of scheduling, the operator employs stricter bounds than those

Table 2

Coefficients of $P = g(Q)$ and minimum operating flow limits.

Pump	a_p	b_p	c_p	$Q_p^{\min}$ (m^3/h)
p_1	0.02	-0.33	10.34	14.1
p_2	0.026	-0.35	8.76	11.5
p_3	0.013	-0.29	9.29	17.7
p_4	0.005	-0.26	21.39	43.5
p_5	0.003	-0.13	10.15	34.2
p_6	0.011	-0.27	15.31	23.4
p_7	0.008	-0.13	6.24	15.1
p_8	0.018	-0.23	6.71	11.7
p_9	0.009	-0.26	13.31	24.9
p_{10}	0.012	-0.29	10.95	19.9
p_{11}	0.014	-0.36	12.48	20.9
p_{12}	0.012	-0.17	5.81	13.2

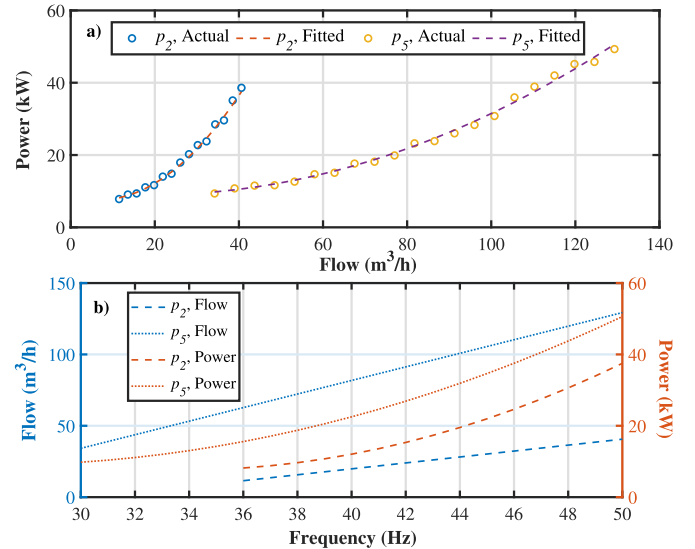


Fig. 7. Pumps p_2 and p_5 : (a) measurement-based process and extraction of quadratic $P = g(Q)$ function; (b) lookup tables with respect to frequency.

Table 3
Accuracy metrics of quadratic fit.

Pump	RMSE (%)	R^2
p_1	1.70	0.9952
p_2	1.65	0.9955
p_3	1.56	0.9960
p_4	2.14	0.9928
p_5	1.85	0.9948
p_6	2.01	0.9938
p_7	2.65	0.9885
p_8	1.87	0.9943
p_9	1.76	0.9952
p_{10}	1.55	0.9964
p_{11}	1.70	0.9955
p_{12}	1.84	0.9946

considered in the proposed framework ($SoC_{\tau}^{\min} = 0.2$ p.u. and $SoC_{\tau}^{\max} = 0.9$ p.u.), in order to reduce the risk of tank overflow or depletion. In addition, Base Case 1 is examined using minimum and maximum SoC bounds of 0.2 p.u. and 0.9 p.u. (Base Case 1b), respectively, to enable a fairer comparison with the proposed approach.

Regarding Base Case 2, it corresponds to the problem of scheduling the pumps to minimize the operating cost of the WDSs under dynamic electricity pricing only, without considering the provision of grid services. Therefore, Base Case 2 is defined by the following day-ahead problem:

$$\min \sum_{t \in \mathcal{T}} \left(\sum_{p \in \mathcal{N}_p} P_p^k[t] \right) \cdot C[t] \cdot \Delta t$$

Base Case 2: s.t. (10), (12)–(13), (16),

for $SoC_{\tau}^{\max} = 0.9$ p.u., $SoC_{\tau}^{\min} = 0.2$ p.u.,

and $SoC_{\tau}^0 = 0.5$ p.u., $\forall \tau \in \mathcal{N}_{\tau}$.

The differences in the operational paradigm between the two base cases and the proposed method are indicated in Fig. 8, where the power consumption and the flow of pump p_9 , as well as the SoC of tank τ_5 , are illustrated. As observed, in both Base Case 1a and Base Case 1b, the pump setpoints are unresponsive to electricity price variations, being determined exclusively by the SoC of the downstream tank. Moreover, Base Case 1b results in longer pump operating cycles, as the tank SoC operating range is wider.

In contrast, in Base Case 2, the pump setpoints are higher during the periods of lower electricity prices. Under the proposed framework, between 00:00 and 18:00, the pump power setpoints are scheduled to vary less than in Base Case 2, to reserve both upward and downward capacity. Moreover, the pump is switched off during the $\mathcal{T}_{I/C-L}$ period to achieve power reduction. Finally, once the $\mathcal{T}_{I/C-L}$ period elapses, the pump setpoints under the proposed framework are higher than those in Base Case 2, to compensate for the prolonged shutdown and to recover the SoC.

In terms of the entire WDS, the total power consumption is presented in Fig. 9, with the value of 624.4 kW corresponding to $\sum_{p \in \mathcal{N}_p} P_p^{\max}$. Regarding costs, in Base Case 1a, the daily operating cost is \$567, whereas in Base Case 1b, the longer pump operating cycles increase this cost to \$607. The increased cost of Base Case 1b compared to Base Case 1a showcases that relaxing SoC operating limits without optimized scheduling neither increases operational flexibility nor guarantees lower operating costs. In contrast, applying the schedule of Base Case 2 reduces the daily operating cost to \$300, i.e., by 47.1% compared to Base Case 1a. For Base Case 1a, Base Case 1b, and Base Case 2, the calculated costs refer to the daily energy cost for the operation of pumps. Finally, by applying the proposed framework and enabling the provision of grid services, the net daily operating cost is further reduced to

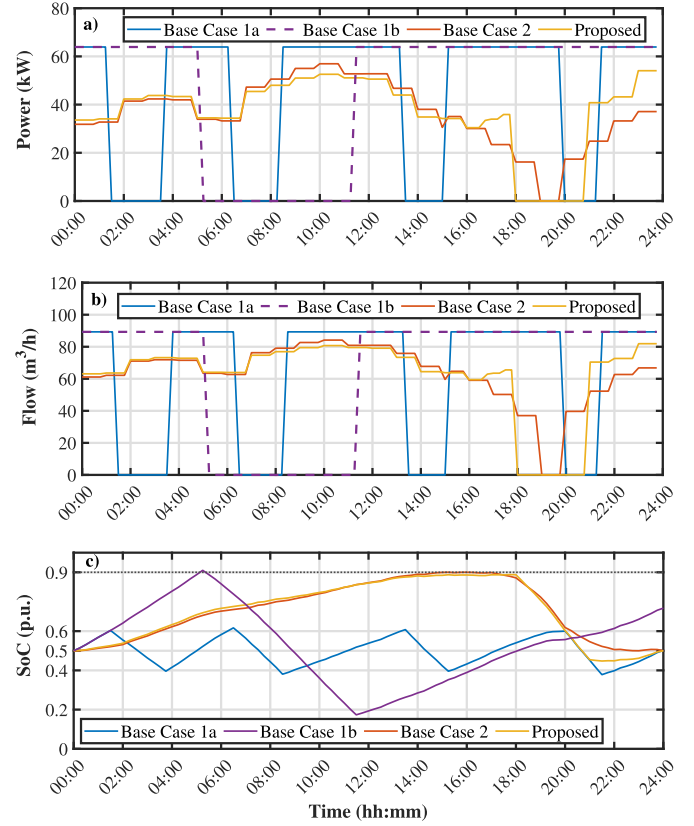


Fig. 8. (a) power and (b) flow of pump p_9 ; (c) SoC of tank τ_5 .

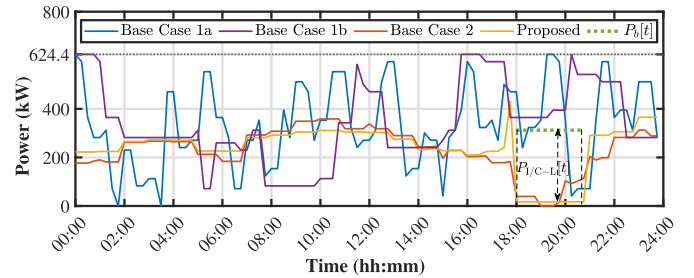


Fig. 9. Total power consumption of the WDS.

\$70, corresponding to an 87.6% reduction compared to Base Case 1a. More specifically, under the proposed framework, the actual pump energy cost is \$314, i.e., slightly higher than in Base Case 2. Nevertheless, the revenues from the I/C-L, upward reserve capacity, and downward reserve capacity services are \$177, \$17, \$50, respectively, thereby reducing the net operating cost to \$70. Therefore, the proposed framework leverages the operational flexibility of WDSs, enabling them to generate significant revenue by providing substantial services to the EPS while also satisfying their operational requirements, such as the continuity of water supply.

To elaborate on the services provided, as shown in Fig. 9, the power reduction achieved for the I/C-L service is 295 kW, sustained over three hours, resulting in a total of 885 kWh curtailed. Moreover, as shown in Fig. 10, for the balancing service, both the upward and the downward reserve capacities exceed 200 kW, $\forall t \in \mathcal{T}_R$. In general, the reservation of downward capacity is prioritized, both due to the higher compensation rate during most time intervals and because it corresponds to lower power consumption, and thus lower operating costs. This explains the

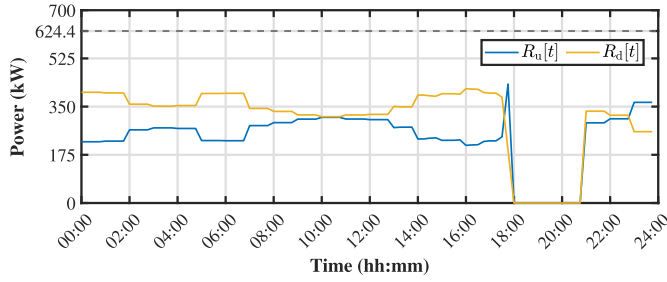


Fig. 10. Upward and downward reserve capacities under the proposed framework.

higher revenue from the provision of downward reserve capacity compared to upward reserve capacity: \$50 and \$17, respectively. Finally, it is noted that during the $T_{I/C-L}$ period, reserve capacities are zero, as the two services do not overlap.

Overall, the consumption and service provision metrics of the WDS across the examined cases are summarized in Table 4. As observed, the optimal scheduling of Base Case 2 and the proposed framework significantly reduces the daily energy consumption of the WDS compared to both Base Case 1a and Base Case 1b, thereby achieving energy savings while ensuring water supply. Furthermore, it is noted that, from the EPS perspective, the $P_{I/C-L}[t]$, $R_d[t]$, and $R_u[t]$ provided by the proposed framework reflect the reduction in peak generation and balancing services required from conventional generation units, and thereby the corresponding reduction in operating costs and CO₂ emissions.

Moreover, to demonstrate the importance of incorporating the reserve activation constraints (14) and (15) into the proposed day-ahead problem, a third base case is examined. To elaborate, Base Case 3 involves the problem of Section 3.2 but without the inclusion of (14) and (15). Specifically:

$$\begin{aligned} \min \quad & (7) \\ \text{s.t.} \quad & (8)-(10), \quad (12)-(13), \quad (16), \\ \text{Base Case 3:} \quad & \text{for } SoC_{\tau}^{\max} = 0.9 \text{ p.u.}, SoC_{\tau}^{\min} = 0.2 \text{ p.u.}, \\ & \text{and } SoC_{\tau}^0 = 0.5 \text{ p.u.}, \forall \tau \in \mathcal{N}_{\tau}^*. \end{aligned}$$

Base Case 3 and the proposed optimization framework are compared in Fig. 11, which illustrates the effect of including constraints (14) and (15) on the tank SoC. To elaborate, as shown in Fig. 11, for tank τ_3 under the proposed framework, the tank SoC, $SoC_{\tau}[t]$, is bounded by the auxiliary variables $SoC_{\tau}^u[t]$ and $SoC_{\tau}^d[t]$, which ensure that the SoC limits will not be violated in the event of reserve activation in the intra-day stage. In contrast, Base Case 3 does not include (14) and (15); thus, it fails to prevent the SoC from repeatedly assuming values at both the upper and lower operating limits, potentially leading to violations in the event of reserve activation in the intra-day stage. In essence, the reserve activation constraints ensure that the reserve capacities nominated

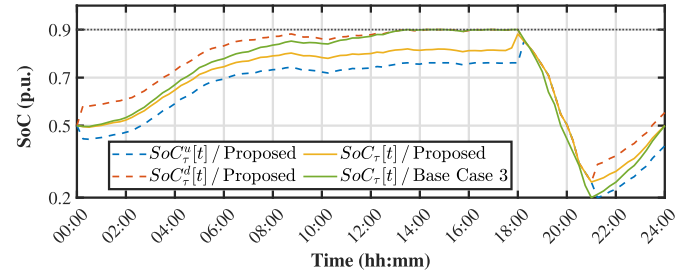


Fig. 11. Tank τ_3 : Profiles of $SoC_{\tau}[t]$, $SoC_{\tau}^d[t]$, and $SoC_{\tau}^u[t]$ in the proposed framework, along with $SoC_{\tau}[t]$ in Base Case 3.

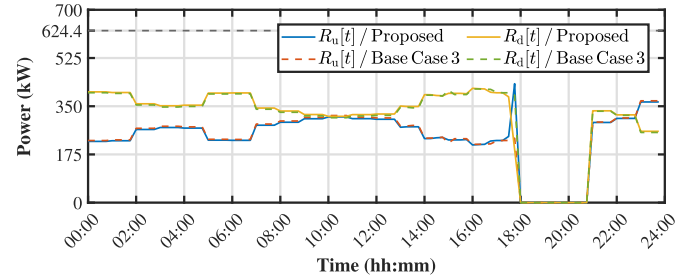


Fig. 12. Upward and downward reserve capacities under the proposed framework and Base Case 3.

to the EPS operator are calculated more conservatively, accounting for the activation stage. The upward and downward reserve capacities for the proposed framework and Base Case 3 are visualized in Fig. 12. The deviations in total reserve capacities between the two examined cases appear negligible at each time interval; however, their cumulative effect over time results in significant differences in the tank SoC profiles, as observed in Fig. 11.

4.4. Validation of the intra-day stage

In this section, the intra-day stage of the proposed framework is validated. As discussed in Section 3.3, the activation request can involve both upward and downward balancing powers and occur at any $t \in T_R$. Without loss of generality, two distinct scenarios are considered: 1) an upward activation request at 21:15; 2) a downward activation request at 15:00. For the penalty weight c_{τ} in (19), the value of 20,000 is used $\forall \tau \in \mathcal{N}_{\tau}^*$. The total WDS power consumption profiles for the two scenarios are presented in Fig. 13(a) and (b). In the same figures, the original profiles, as derived by the day-ahead problem, are also plotted. As observed, apart from the reserve activation, the intra-day problem reschedules the pump operating setpoints to minimize the actual operating cost for the remainder of the day and ensure SoC recovery by the end of the day. Indicatively, for the downward activation scenario of Fig. 13(b), rescheduling the pump setpoints achieves an additional 10% reduction in operating cost compared to the original day-ahead schedule, specifically from \$100 to \$90. Regarding SoC recovery, the effectiveness of the intra-day problem is demonstrated in Fig. 14. As shown, if the pump schedule is not adjusted and the original day-ahead schedule is followed instead, the SoC is not recovered to its initial value.

Finally, in addition to the analysis of Section 4.3, the importance of the reserve activation constraints (14) and (15) is further verified during the intra-day stage. As discussed in Section 4.3 and demonstrated in Fig. 11, the absence of constraints (14) and (15) in Base Case 3 can result in SoC violations in the event of reserve activation during the intra-day stage. This is illustrated in Fig. 15. As observed, for Base Case 3, both the upward and downward activation scenarios cause violations of the SoC

Table 4

Operational metrics of the WDS. For the base cases, only consumption metrics are included, as grid services are not provided.

	Base Case			Proposed
	1a	1b	2	
Peak power consumption (kW)	624.4	624.4	358.4	431.8
Average power consumption (kW)	332.3	343.2	231.9	239.5
Daily energy consumption (kWh)	7975.2	8236.8	5565.6	5748
Power reduction $P_{I/C-L}[t]$ (kW)	-	-	-	295
Peak $R_u[t]$ (kW)	-	-	-	431.8
Peak $R_d[t]$ (kW)	-	-	-	415
Average $R_u[t]$ (kW)	-	-	-	271.4
Average $R_d[t]$ (kW)	-	-	-	353

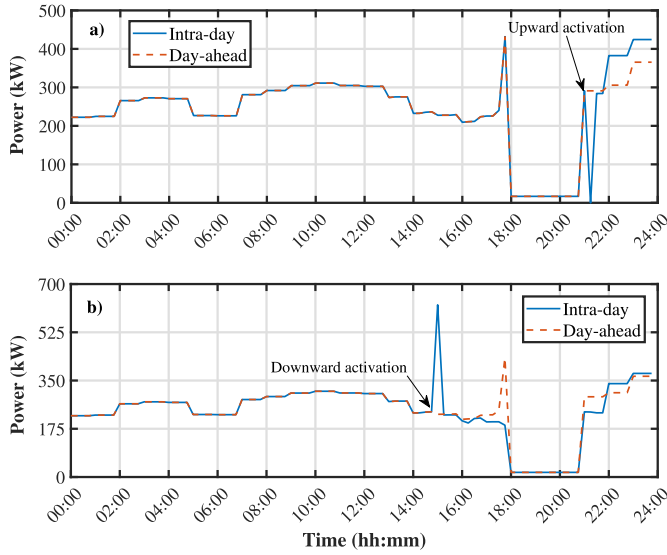


Fig. 13. WDS total power consumption according to the day-ahead schedule and its intra-day adjustment, for (a) upward and (b) downward activation.

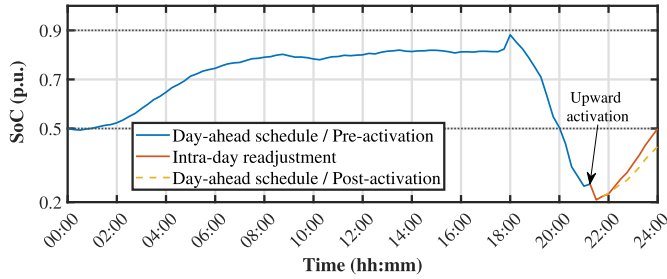


Fig. 14. Tank τ_3 : SoC profile at upward request. The blue and yellow lines depict the day-ahead schedule, before and after the activation interval, respectively. The red line depicts the intra-day adjustment. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

limits for tank τ_3 . In contrast, day-ahead and the intra-day schedules of the proposed framework ensure that limits are respected.

4.5. Impact of balancing capacity quantification

In this section, different approaches for defining and quantifying the balancing capacity of WDSs are compared and discussed with respect to their impact on the provision of the service and the economic benefits of WDSs. The following approaches are considered:

Approach 1: Single, symmetric balancing capacity for both upward and downward service provision, quantified on a per-pump basis as per (21)–(22).

$$R_p[t] = \min\left(P_p^k[t], P_p^{\max} - P_p^k[t]\right), \quad p \in \mathcal{N}_p \quad (21)$$

$$R_1[t] = \sum_{p \in \mathcal{N}_p} R_p[t]. \quad (22)$$

In (21), $R_p[t]$ is the single, symmetric capacity of each pump for both upward and downward services. In (22), $R_1[t]$ denotes the single, symmetric capacity of the entire WDS for Approach 1. This approach is adopted by the study of [21].

Approach 2: Single, symmetric balancing capacity for both upward and downward service provision, quantified on a system-wide basis as per (23). The balancing capacity of the WDS for this approach is denoted

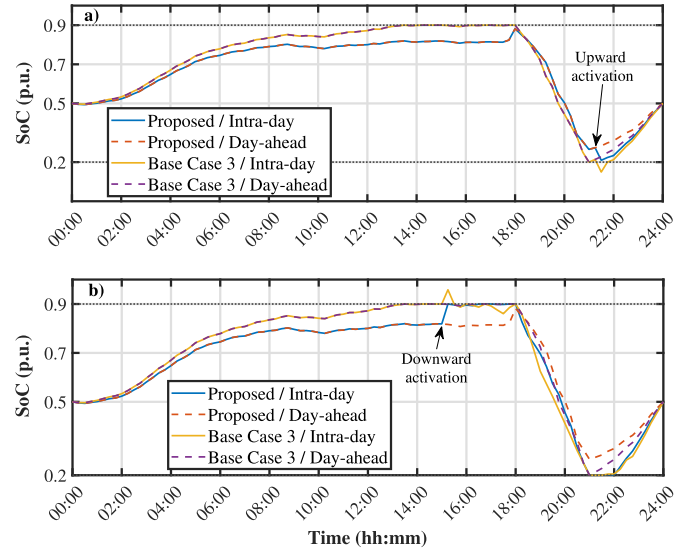


Fig. 15. Tank τ_3 : SoC profile according to the day-ahead schedule and its intra-day adjustment, for (a) upward and (b) downward activation, in the proposed framework and Base Case 3.

as $R_2[t]$.

$$R_2[t] = \min\left(\sum_{p \in \mathcal{N}_p} P_p^k[t], \sum_{p \in \mathcal{N}_p} (P_p^{\max} - P_p^k[t])\right). \quad (23)$$

Proposed approach: Distinct balancing capacities for upward and downward services, quantified on a system-wide basis as per (9).

The three approaches are compared in Fig. 16, which depicts the profiles of $R_1[t]$, $R_2[t]$, $R_u[t]$, and $R_d[t]$ for the WDS under study and the parameters of Section 4.1. For the proposed approach, the compensation rates of Fig. 5(b) are used. For Approach 1 and Approach 2, the symmetric balancing capacity is compensated with the sum of the upward and downward compensation rates.

The results of Fig. 16 illustrate the differences in the mathematical quantification of balancing capacities. Specifically, the proposed approach allows for distinct upward and downward capacities, the optimal values of which are calculated according to the different compensation rates. In contrast, as observed in Fig. 16, the profile of $R_2[t]$ is similar to that of $R_u[t]$ and lower than that of $R_d[t]$. This is because Approach 2 limits the capability of the WDS for balancing service provision to the minimum of the margins for increasing or decreasing power consumption, as shown in (23). Moreover, as shown in (22), this capability is further restricted in Approach 1, which quantifies the total WDS capacity as the sum of the locally limited capacities of individual pumps. For

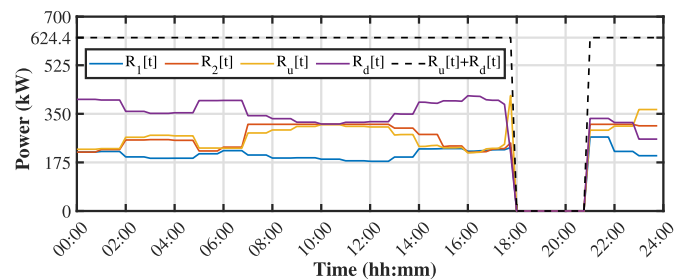


Fig. 16. Balancing capacities for approach 1 ($R_1[t]$), approach 2 ($R_2[t]$), and the proposed approach ($R_u[t]$, $R_d[t]$).

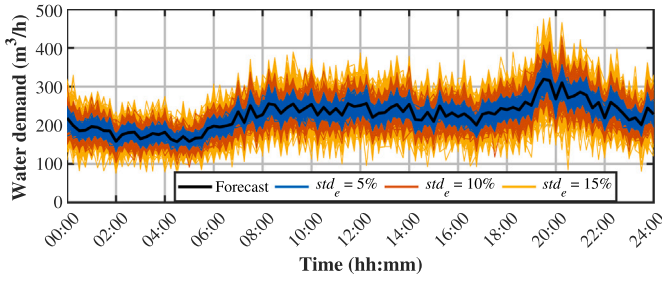


Fig. 17. Tank τ_5 : Forecast aggregate water demand and 1000 MC realizations for distinct std_e values.

this reason, $R_1[t]$ is generally lower than $R_2[t]$ in Fig. 16, visualizing the importance of system-wide quantification of the balancing capacity.

In summary, the proposed approach, which introduces distinct, system-wide upward and downward capacities, is the most beneficial for both the EPS and WDS operators. For the EPS operator, it ensures that balancing services are tailored to system conditions, as reflected in the different upward and downward compensation rates. For the WDS, it ensures that flexibility is not unnecessarily restricted. The economic superiority of the proposed approach is reflected in the net daily operating costs of the WDS, which are \$89, \$76, and \$70 for Approach 1, Approach 2, and the proposed approach, respectively. Thus, the net daily operating cost is reduced by 21.3% under the proposed approach compared to the approach of [21] (Approach 1).

4.6. Impact of water demand uncertainty

Here, the impact of water demand uncertainty is examined. To elaborate, in the results of Sections 4.3 and 4.4, water demand forecasts are assumed to be perfectly accurate; however, in practice, forecasts are subject to errors.

More specifically, this section examines the impact of water demand uncertainty and forecast errors on tank water levels. The analysis focuses on the indicative scenarios of Section 4.4, i.e., intra-day pump schedules with upward reserve activation at 21:15 (Scenario 1) or downward reserve activation at 15:00 (Scenario 2). Forecast errors are considered Gaussian [15], with zero mean value and standard deviation std_e . Three std_e levels are considered, namely, 5%, 10%, and 15% of the forecast value. For each error level, 1000 Monte Carlo (MC) realizations are generated.

The analysis investigates the percentages of tank overflow or depletion events for Scenario 1 and Scenario 2 across the 1000 MC realizations for the three error levels. An MC realization is considered to cause overflow or depletion when the SoC of at least one tank reaches 1 p.u. or 0 p.u., respectively, at any time during the day. For each scenario, the pump schedule remains fixed across all MC realizations and corresponds to the schedule obtained via the day-ahead problem and adjusted via the intra-day problem, as presented in Fig. 13. Indicatively, Fig. 17 depicts the forecast aggregate water demand to be supplied by tank τ_5 , along with the 1000 MC realizations for the three error levels.

Table 5 reports the percentages of overflow and depletion incidents across the 1000 MC realizations for the two scenarios and the three error levels. As shown, for both scenarios, the proposed optimization framework remains robust against errors with $std_e = 5\%$, as zero incidents of either depletion or overflow are reported across all 1000 MC cases. This robustness is attributed to two factors. First, the optimization framework imposes conservative SoC operational limits, namely $SoC_{\tau}^{\max} = 0.9$ p.u. and $SoC_{\tau}^{\min} = 0.2$ p.u., instead of the physical bounds of 1 p.u. and 0 p.u., thereby partially hedging against forecast uncertainty. Second, forecast errors tend to partially cancel out over the daily time horizon.

For $std_e = 10\%$, depletions are still prevented, while overflows occur in only 2.9% and 4.7% of the examined cases for Scenario 1 and

Table 5

Percentages of overflow or depletion incidents for Scenario 1 and Scenario 2 in 1000 MC realizations under different error levels.

Incident	$std_e = 5\%$		$std_e = 10\%$		$std_e = 15\%$	
	Scenario 1	Scenario 2	Scenario 1	Scenario 2	Scenario 1	Scenario 2
Depletion	0	0	0	0	1.2	1.7
Overflow	0	0	2.9	4.7	16.8	28.1

Scenario 2, respectively. For $std_e = 15\%$, the percentage of overflows increases for both scenarios. Nevertheless, depletion incidents, which pose the most critical risk for WDS operation as they correspond to water supply interruptions, remain limited to 1.2% and 1.7% for Scenario 1 and Scenario 2, respectively. Overall, overflows are more frequent in Scenario 2 because the downward reserve activation pushes SoC profiles closer to 0.9 p.u. for extended time periods, as observed in Fig. 15. Moreover, for both scenarios, overflows occur more frequently than depletions because the enforced operating limits $SoC_{\tau}^{\max}$ and $SoC_{\tau}^{\min}$ are asymmetric, with $SoC_{\tau}^{\min}$ being stricter. This asymmetry is intentionally introduced into the proposed optimization framework to prioritize the prevention of depletions, which are more critical for the WDS operation.

In summary, although water demand uncertainty is not explicitly incorporated into the optimization framework, the WDS schedule remains robust against forecast errors of $std_e = 5\%$ and exhibits satisfactory performance for forecast errors of $std_e = 10\%$ and $std_e = 15\%$.

4.7. Discussion

The results of this section demonstrate both the accuracy of the proposed modeling approach and the effectiveness of the proposed optimization framework in enabling WDSs to provide multiple grid services. The accuracy of the proposed modeling approach ensures that the schedules derived by the proposed optimization framework are consistent with the physical laws governing the WDS. Furthermore, from an economic perspective, the results highlight the substantial benefits for WDSs and underscore the importance of adjusting the pump schedule during the intra-day stage. From a technical perspective, the results emphasize the significance of incorporating the reserve activation constraints.

Moreover, it is emphasized that the development of the proposed measurement-based modeling approach not only addresses the research drawbacks identified in Section 2.2; it also tackles the practical problem of modeling the examined real-world WDS. Specifically, the system curves of the Igoumenitsa WDS are unknown to its operator. As a result, conventional methods [8–14,16–21,24,25], which rely on the modeling assumptions outlined in Section 2.2, cannot be applied, necessitating the development of a measurement-based approach.

Regarding the convex relaxation of constraint (11), as explained in Section 3.2, in the vast majority of cases, the structure of the optimization problem ensures that the relaxed constraint (16) is satisfied with equality, thereby preserving the equivalence between the original and the relaxed problems. Specifically, for the examined cases, the numerical discrepancy between the calculated power setpoints and the corresponding quadratic flow terms in (16) is of the order on 10^{-7} kW, demonstrating exactness. In turn, this convexification achieves a significant reduction in the computation time. Indicatively, the day-ahead problem of Section 4.3 includes 15,604 constraints and 11,993 variables, 1152 of which are binary. For this problem, the convex relaxation decreases the computation time from 41 s to 5 s, i.e., by 88%. This improvement in computation time is particularly important for large-scale WDSs comprising numerous pumps and tanks, or for applications where a small time step is required.

On the other hand, the main limitation of the study is that water demand uncertainties are not systematically accounted for. As analyzed in Section 4.6, for Gaussian forecast errors with std_e up to 5%, the proposed framework ensures satisfaction of the WDS technical limits. Therefore, systematically hedging against higher levels of uncertainty

remains a task for future research. Possible directions include extending the proposed optimization framework to a stochastic formulation [13,19], or employing reinforcement learning [49–51] or data-enabled predictive control techniques [52,53] for corrective, real-time scheduling. Finally, as an additional direction for future research, the proposed measurement-based modeling approach should be extended to WDSs with configurations that allow direct pumping to water consumers.

5. Conclusions

This paper proposes a practical, two-fold methodology to enable the provision of multiple grid services by WDSs. The methodology comprises two parts: a new measurement-based approach for modeling WDSs, and a new two-stage optimization framework for scheduling WDSs, encompassing the day-ahead and the intra-day time horizons.

The proposed measurement-based modeling approach addresses the main drawbacks of the current modeling practices for WDSs, particularly the dependence on parameter knowledge and the complexity of the associated optimization problem formulations for scheduling WDSs. To elaborate, the proposed modeling approach involves an on-site measurement-based process that correlates the electric power consumption and the flow of VSPs. This approach ensures more accurate modeling, as it relies on up-to-date measurements, while it decouples power and flow from the remaining physical quantities, such as hydraulic head, and pump efficiency speed. It is shown that the electric power and the flow of a VSP can be accurately correlated through a convex quadratic function, enabling simpler formulations for the associated optimization problems.

Subsequently, by leveraging the above-mentioned modeling approach, the proposed two-stage optimization framework addresses the main limitations of the existing frameworks, enabling the provision of multiple grid services by WDSs. More specifically, I/C-L and balancing services are examined, along with dynamic electricity pricing. The day-ahead optimization problem derives the initial pump schedule to minimize the net daily operating cost of the WDS, ensuring stable power reduction for the I/C-L service, as well as upward and downward reserve capacities for the balancing service. Moreover, contrary to the majority of the existing literature, the day-ahead problem ensures that the reserve activation during the intra-day stage does not violate the tank SoC limits. Regarding the intra-day optimization problem, it determines the optimal pump setpoints for reserve activation and adjusts the pump schedule for the remainder of the day to minimize the operating cost, while also ensuring the recovery of the tank SoC. Both the day-ahead and the intra-day problems are formulated as MIQP problems with convex continuous relaxations, supporting tractable solutions.

The proposed methodology was applied and validated on a real-world WDS located in northwestern Greece. First, the proposed measurement-based modeling approach was performed and the quadratic curves for the WDS pumps were derived. Subsequently, the proposed optimization framework for scheduling WDSs was validated, using real data for water demands. The results demonstrate that the proposed methodology unlocks the flexibility of WDSs and enables a reduction of nearly 90% in their net daily operating cost, compared to the current operational practices.

CRedit authorship contribution statement

Stelios Dimoulas: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Georgios Kryonidis:** Writing – review & editing, Supervision. **Kyriaki-Nefeli Malamaki:** Writing – review & editing, Supervision. **Pandelis Biskas:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

In this appendix, the proposed optimization framework is: a) first generalized to accommodate FSPs, and b) subsequently applied to a benchmark WDS.

A.1. Generalization for FSPs

For WDSs that include both VSPs and FSPs, the set of pumps $\mathcal{N}_p$ is defined as $\mathcal{N}_p = \mathcal{N}_p^{\text{VSP}} \cup \mathcal{N}_p^{\text{FSP}}$, where $\mathcal{N}_p^{\text{VSP}}$ and $\mathcal{N}_p^{\text{FSP}}$ are disjoint subsets that represent the VSPs and FSPs of the WDS, respectively.

Therefore, constraints (11)–(12) and (13b)–(13c) are reformulated as:

$$P_p[t] = a_p \cdot Q_p^2[t] + b_p \cdot Q_p[t] + c_p, \quad p \in \mathcal{N}_p^{\text{VSP}}, t \in \mathcal{T}. \quad (24)$$

$$P_p^k[t] \leq P_p[t] - P_p^{\min} \cdot (1 - k_p[t]) \quad (25a)$$

$$P_p^k[t] \geq P_p[t] - P_p^{\max} \cdot (1 - k_p[t]) \quad (25b)$$

$$P_p^k[t] \leq P_p^{\max} \cdot k_p[t] \quad (25c)$$

$$P_p^k[t] \geq P_p^{\min} \cdot k_p[t] \quad (25d)$$

$$Q_p^k[t] \leq Q_p[t] - Q_p^{\min} \cdot (1 - k_p[t]) \quad (25e)$$

$$Q_p^k[t] \geq Q_p[t] - Q_p^{\max} \cdot (1 - k_p[t]) \quad (25f)$$

$$Q_p^k[t] \leq Q_p^{\max} \cdot k_p[t] \quad (25g)$$

$$Q_p^k[t] \geq Q_p^{\min} \cdot k_p[t] \quad p \in \mathcal{N}_p^{\text{VSP}}, t \in \mathcal{T}. \quad (25h)$$

$$P_p^{\min} \leq P_p[t] \leq P_p^{\max}, \quad p \in \mathcal{N}_p^{\text{VSP}}, t \in \mathcal{T} \quad (26a)$$

$$Q_p^{\min} \leq Q_p[t] \leq Q_p^{\max}, \quad p \in \mathcal{N}_p^{\text{FSP}}, t \in \mathcal{T}. \quad (26b)$$

To model FSPs, equation (27) needs to be incorporated into the optimization framework, for $p \in \mathcal{N}_p^{\text{FSP}}, t \in \mathcal{T}$.

$$P_p^k[t] = P_p^{\max} \cdot k_p[t]; \quad Q_p^k[t] = Q_p^{\max} \cdot k_p[t]. \quad (27)$$

Specifically, since FSPs do not operate under different input frequencies, continuous decision variables $P_p[t]$ and $Q_p[t]$ cannot be defined. Instead, pump electric power consumption and flow are either zero when the pump is shut down (binary variable $k_p[t] = 0$), or equal to the operating points $P_p^{\max}$ and $Q_p^{\max}$, measured at the nominal frequency f_n , when the pump is running (binary variable $k_p[t] = 1$).

A.2. Application on benchmark WDS

In order to test the scalability and generality of the proposed optimization framework, this section examines its application to a benchmark WDS. Specifically, the proposed framework is applied to the topology of the Richmond benchmark WDS, as illustrated in Fig. 18 and described in [54]. As observed, the WDS is supplied by a single water source and comprises six storage tanks and seven pumps. Specifically, the main tank of the WDS is tank τ_{1r} , which is directly supplied by the water source via the combined operation of a pumping station with two pumps and a separate booster pump. In turn, tank τ_{1r} supplies tanks τ_{2r} – τ_{6r} via additional booster pumps.

In line with Appendix A.1, both VSPs and FSPs are considered. Specifically, the pumps that supply τ_{1r} are treated as FSPs (pumps $p_{1,\text{FSP}}, p_{2,\text{FSP}}, p_{3,\text{FSP}}$) and are assumed to be switched on and off simultaneously, since the WDS topology implies that they should operate together at nominal speed to effectively extract and transfer water. The remaining booster pumps are considered VSPs. Given that the proposed optimization framework is applied theoretically, i.e., without the prior application of the proposed measurement-based modeling approach, theoretical parameters are considered for the tanks and pumps. For the

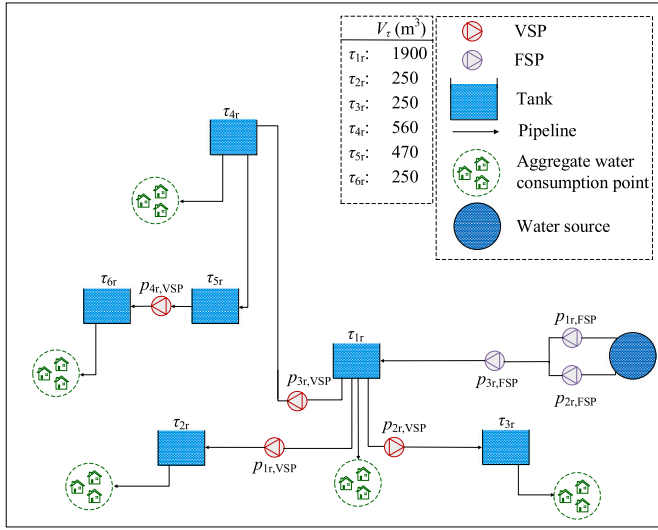


Fig. 18. Schematic of the Richmond benchmark WDS.

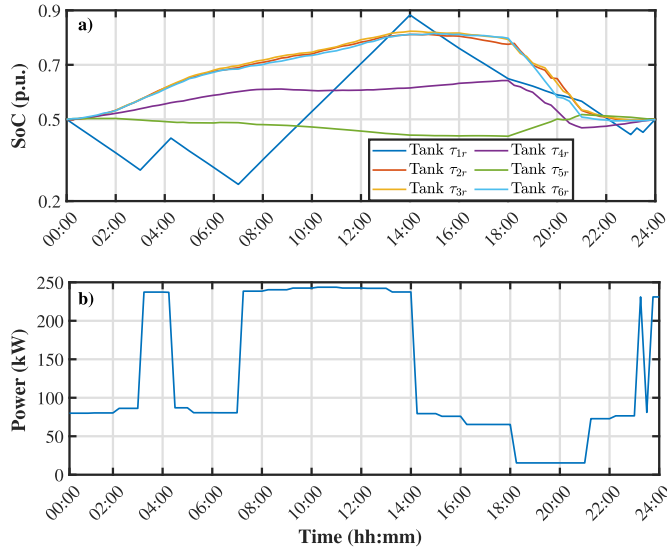


Fig. 19. a) SoC profiles; b) total WDS power consumption.

tanks, the assumed volumes are listed in Fig. 18. The three FSPs are assumed to operate together as an equivalent pump with $p_p^{\max} = 150$ kW and $Q_p^{\max} = 300$ m³/h. For VSPs $p_{1r,VSP}$, $p_{2r,VSP}$, $p_{3r,VSP}$, and $p_{4r,VSP}$, the parameters of pumps p_8 , p_{12} , p_1 , and p_2 of the Igoumenitsa WDS are used, respectively. Finally, without loss of generality, all aggregate water consumption points are assumed to follow the water demand profile of tank τ_4 of the Igoumenitsa WDS, as depicted in Fig. 5(a). For the electricity cost and the service compensation rates, the prices of Fig. 5(b) are considered.

The tank SoC profiles and the total WDS power consumption are illustrated in Fig. 19(a) and (b), respectively. The net daily operating cost of the WDS is \$ 97. The total execution time is 6 s, demonstrating the scalability and applicability of the proposed optimization framework to WDSs with interconnected storage tanks.

Data availability

Data will be made available on request.

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